

Uncertainty-Aware and Grammar-Constrained Structured Decoding for Library Call Number Recognition

Chengyue Shen

High School Student

Shanghai Foreign Language School Affiliated to SISU

Shanghai, China

shenchengyue8@gmail.com

Abstract—Manual library inventory is labor-intensive, while OCR often confuses similar characters, separates multiline call numbers, and retains irrelevant text. We propose a context-aware conditional random field (CRF) for recognizing Chinese Library Classification call numbers in shelf images. It preserves character alternatives, jointly handles character correction, region grouping, and background text elimination under grammar constraints, while penalizing conflicting leading classification letters at neighboring sites. The dataset contains 242 images and 3,907 call numbers from three libraries. The model achieved 94.59% F1 on 48 test images containing 815 call numbers, compared with 85.90% for deterministic post-processing method. Trained on the first two libraries and tested on the third, it achieved 90.49% F1 on 965 call numbers. Ablations support retaining character alternatives, flexible spans, region connection features, and context factor.

Index Terms—optical character recognition, conditional random fields, grammar-constrained decoding, structured text recognition, library call number recognition

I. INTRODUCTION

Library inventory checks require staff to repeatedly inspect call numbers, which are short alphanumeric codes printed on book spines for identification. While the process is labor-intensive and prone to human error, optical character recognition (OCR) can reduce this manual work by reading shelf images automatically.

While general-purpose OCR systems such as PaddleOCR [1] perform well on plain, unstructured text, they produce systematic errors that are difficult to fix with generic post-processing pipelines when dealing with structured text such as library call numbers, vehicle identification numbers, or product serial codes. Similar characters are easily misread (1 vs. l), one call number may be split into different OCR regions, and irrelevant text may be included incorrectly. Once a conventional OCR decoder has fixed to one preferred string for each region, later processing loses alternative character evidence that could resolve these errors.

Prior camera-based library systems have combined text and visual information to identify book spines [2] and have applied scene-text reading to inventory construction and retrieval [3]. More recently, Llabrés et al. combined book-spine segmentation, OCR, and many-to-many matching against a catalog

[4]. Related scene-text methods have consolidated character evidence, spatial constraints, and linguistic information in a CRF [5], or grouped character candidates into text lines using min-cost flow [6]. STEP uses regular expressions to guide the detection and recognition of structured scene text [7]. The library-oriented systems above identify complete books from spine text, catalog metadata, or visual appearance. The structured text methods address related OCR problems, but do not jointly model character uncertainty, fragmented call number regions, and call number grammar.

This study addresses this gap by extracting Chinese Library Classification (CLC) call numbers from bookshelf spine images. The method retains multiple candidate characters at each position and constructs interpretations from nearby OCR regions. A deterministic finite automaton (DFA) discards strings that violate the call-number grammar. A CRF scores the remaining candidates using OCR confidence and learned structural features. A shelf-context factor penalizes conflicting leading CLC letters at neighboring call number sites, and global maximum-a-posteriori (MAP) yields an output with the highest score.

The contributions of this study are:

- a joint framework for character correction, region grouping, separator insertion, background rejection, and shelf context;
- an annotated dataset containing 3,907 call numbers from 242 shelf images collected at three libraries; and
- an evaluation showing 94.59% F1 compared with 84.90% F1 for deterministic post-processing.

These results show the value of retaining OCR uncertainty and solving dependent decisions jointly. Its modular grammar and scoring model may facilitate adaptation to other structured-text tasks, although such transfer requires retraining and further evaluation.

The remainder of this paper is organized as follows. Section II presents the proposed method, Section III reports the experiments, and Section IV concludes the paper.

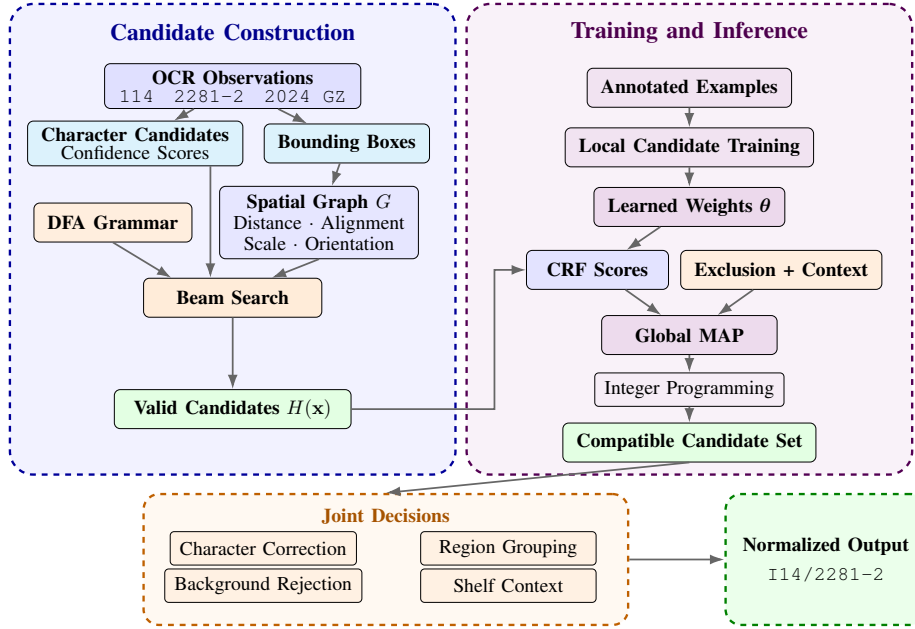


Fig. 1. Framework of the proposed method. The context factor adds a penalty for conflicting leading CLC letters at neighboring sites.

II. METHOD

Figure 1 summarizes the method. It retains multiple character candidates and their OCR scores, constructs a spatial graph over detected text regions, and uses a DFA to eliminate interpretations that violate the call-number grammar. Beam search limits the number of partial interpretations. A CRF assigns learned scores to the remaining candidates; inspired by piecewise training [8], its weights are estimated from small groups of competing local interpretations. A shelf-context factor scores pairs of candidates at neighboring sites. Global MAP decoding [9] then selects a compatible set without reusing an OCR region. Character correction, multiline grouping, separator insertion, background rejection, and shelf context are thus coordinated in one structured prediction problem.

A. Task Definition and Input Representation

An OCR system first detects image regions that may contain text and then recognizes the characters in each region. For an image with M detected regions, the evidence retained by this model is

$$\mathbf{x} = \{(b_j, Q_j)\}_{j=1}^M, \quad (1)$$

where b_j is the bounding rectangle of region j , and $Q_j(i, c)$ is the recognizer score for character c at ordered position i . A conventional decoder keeps only the highest-scoring choice at each position and returns one preferred string, thereby discarding alternatives that may become plausible in context.

The evidence in Eq. 1 is obtained using PaddleOCR [1]. Its detector directly supplies b_j , but its conventional interface returns only a preferred string and a region-level score. We therefore read the recognizer output before final decoding and retain several character candidates and their scores at each position to form Q_j .

The task is to infer every call number in an image, correct its characters, and group its OCR regions. A *candidate* is one complete normalized interpretation of a possible call number, including its character choices and supporting regions. Every output has a classification field and a numeric sequence field joined by /, whether they are printed on one or two lines; internal spaces are removed. Labels may also contain an auxiliary code such as 2024 GZ, consisting of a year and library-site abbreviation, which is not part of the call number and must be excluded from the output. This code is referred to as library information in the subsequent text.

B. Model Assumptions

- **Assumption 1:** One OCR region contains at most one call number and belongs to at most one output, although it may also contain an auxiliary library code.
Justification: Two call numbers rarely merge into one region; exclusive assignment prevents fragment reuse.
- **Assumption 2:** OCR preserves character order, and a call number occupies one continuous interval in each region.
Justification: Order errors are rare, and call-number text is printed contiguously.

The model uses only OCR character scores and bounding-box geometry; it does not use an external catalog, label color, or book spine boundary information.

C. Model Formulation

The model constructs grammar-valid candidates, assigns each a learned score, and selects a non-overlapping set. The following parts define the grammar, candidate structure, and CRF score in that order.

1) *Grammar*: The target labels follow the fifth edition of the *Chinese Library Classification* (CLC) [10]. The DFA accepts the verified one- and two-letter prefixes, an optional composite class after :, one normalized /, and numeric suffixes beginning with -, #, or *. Beam search discards a partial string once it cannot reach an accepting DFA state. The grammar rejects a complete auxiliary string such as 2024 GZ, but it cannot prevent a numeric substring from being joined to another fragment; the learned score must resolve such cases.

2) *Candidate Representation*: The OCR regions form a sparse directed graph G whose edges connect geometrically plausible consecutive fragments. A candidate h contains an ordered path through G , one continuous span from each region, a retained character or deletion at each position in that span, and either concatenation or / at each boundary. A before-inside-after state sequence enforces span continuity. Whitespace is removed, and only strings accepted by the DFA are retained. This variable-length representation follows the segmentation idea of semi-Markov CRFs [11] and related component-grouping work in text recognition [12].

3) *CRF Scoring*: A CRF predicts related output variables after the input has been observed [13]. Here, the variables indicate which candidates are selected; shared OCR regions and neighboring call-number sites create dependencies. Each accepted candidate receives the relative score

$$w_{\theta}(h; \mathbf{x}) = \boldsymbol{\theta}^{\top} \mathbf{f}(h, \mathbf{x}). \quad (2)$$

where $\mathbf{f}$ contains OCR scores and ranks, rotation confidence, span and deletion decisions, joins, output form, and DFA transitions. The learned vector $\boldsymbol{\theta}$ weights this evidence; a positive score favors the candidate over leaving its regions as background.

Let $H(\mathbf{x})$ be the candidate pool, S a proposed output set, and $R(h)$ the regions used by h . Without shelf context, the candidate CRF is

$$P_{\theta}(S | \mathbf{x}) = \frac{\exp[\sum_{h \in S} w_{\theta}(h; \mathbf{x})]}{Z_{\theta}(\mathbf{x})} \times \mathbf{1}_{\left[\begin{array}{c} R(h) \cap R(h') = \emptyset \\ \forall h \neq h' \in S \end{array} \right]}. \quad (3)$$

Here, $Z_{\theta}(\mathbf{x})$ normalizes the distribution over compatible output sets. Each candidate has one learned score and a binary selection variable indicating whether it is included in the final output. One region cannot be used in two outputs.

The shelf context factor connects nearby suspected call number sites. Let $a(h)$ be the site of candidate h , $c_1(h)$ its leading CLC letter, d_{pq} the normalized distance between sites p and q , and D_p the degree of site p . For candidates at adjacent sites $p = a(h)$ and $q = a(h')$, the conflict feature is

$$\psi(h, h') = \mathbf{1}_{[c_1(h) \neq c_1(h')]} \frac{\exp(-d_{pq}/\rho)}{\max(D_p, D_q, 1)}, \quad (4)$$

where ρ controls the distance decay. The score of an output set can be expressed as

$$F_{\theta, \eta}(S; \mathbf{x}) = \sum_{h \in S} w_{\theta}(h; \mathbf{x}) + \eta \sum_{\substack{(h, h') \in P \\ h, h' \in S}} \psi(h, h'), \quad (5)$$

with $\eta \leq 0$. The context-aware CRF assigns probability proportional to $\exp F_{\theta, \eta}$ for an output set.

D. Training

The annotations give the correct string, ordered regions, retained spans, character alignments, deletions, and separators; unassigned regions provide background examples. Training all outputs in an image jointly would require summing over every non-overlapping candidate set. Piecewise training avoids this cost by learning from small, explicitly enumerated choices [8]. For neighborhood ℓ , C_{ℓ} contains the correct candidate h_{ℓ}^* , a null option that leaves the neighborhood as background, and plausible incorrect candidates. The objective is

$$\max_{\boldsymbol{\theta}} \sum_{\ell} \left[w_{\theta}(h_{\ell}^*; \mathbf{x}_{\ell}) - \log \sum_{h \in C_{\ell}} e^{w_{\theta}(h; \mathbf{x}_{\ell})} \right] - \frac{\lambda_{\text{reg}}}{2} \|\boldsymbol{\theta}\|_2^2. \quad (6)$$

This learns local feature weights without computing the global partition function.

E. Inference

Inference first generates grammar-valid candidates and then selects a compatible subset by maximum a posteriori (MAP) decoding.

1) *Local Candidate Generation and Decoding*: For each pair of regions, the graph compares normalized distance, alignment, scale, and orientation. Pairs within thresholds are retained. Beam search then traverses G while extending the span state, character choice, separator choice, and DFA state. Invalid partial strings are removed, and the remaining accepted paths form $H(\mathbf{x})$. Because beam search is approximate, a correct path may be removed before final scoring.

2) *Global MAP Decoding*: Let $u_h = 1$ when candidate h is selected, and $z_{hh'} = 1$ when both members of a neighboring pair are selected. Global MAP decoding solves

$$\max_{\{u_h\}, \{z_{hh'}\}} \sum_{h \in H(\mathbf{x})} u_h w_{\theta}(h; \mathbf{x}) + \eta \sum_{(h, h') \in P} z_{hh'} \psi(h, h'), \quad (7)$$

$$\begin{aligned} \text{subject to} \quad & \sum_{h: j \in R(h)} u_h \leq 1, \quad j = 1, \dots, M, \\ & z_{hh'} = u_h u_{h'}, \quad (h, h') \in P, \\ & u_h, z_{hh'} \in \{0, 1\}. \end{aligned}$$

This optimization task is solved with an integer programming solver.

III. EXPERIMENTS

This section compares the complete context-aware model with raw OCR and deterministic post-processing, tests four components through ablation, and evaluates transfer to a third library.

A. Dataset and Experimental Setup

The dataset contains 242 shelf images and 3,907 annotated call numbers from three libraries: respectively 142/2,265, 40/677, and 60/965 images/call numbers. Protocol A splits all 242 images, creating 194 images (3,092 call numbers) for training set and 48 (815 call numbers) for testing set. Protocol B trains on all 182 images from the first two libraries and tests on all 60 images from the third library.

PaddleOCR 3.4.0 [1] was run locally at four rotations. Candidate generation retains ten character alternatives per position, the two strongest rotations per region, local/path beam widths of 32/64, at most 20,000 candidates per image, paths of at most three regions, and four neighbors per region. For shelf context, each call number site is connected to its two nearest sites within 0.2 image diagonals, and $\rho = 0.08$. Piecewise training uses Adam, learning rate 0.03, L_2 strength 0.02, at most 500 epochs, and validation-based early stopping. The context penalty magnitude is selected from $\{0, 0.5, \dots, 3\}$ on the validation images and held fixed for ablations; the selected values are 2.0 for A and 1.5 for B.

Section III-B uses a deterministic post-processing pipeline for comparison. The input of the post-processing baseline involves only conventional top-1 text, whole region confidence, and box geometry. It normalizes case and punctuation, removes auxiliary library information, and extracts alphanumeric runs. It then applies fixed substitutions for common confusions such as $\text{I}/1$ and $\text{O}/0$, and pairs two regions until all codes satisfy grammar. A pair must satisfy fixed distance, alignment, scale, and orientation thresholds.

Evaluation includes every annotated call number, even when its exact path is absent from the retained OCR evidence. Predictions and annotations are compared as multisets, preserving repeated call numbers. An exact match is a true positive (TP); remaining predictions and annotations are false positives (FP) and false negatives (FN). F1 score is defined accordingly. We report precision, recall, F1, and images whose entire output is correct.

B. Comparison Experiments

Two baselines and our model are evaluated using Protocol A. *Raw OCR* returns the recognized string with highest confidence from every detected OCR region after deleting whitespace. *OCR plus post-processing* applies the deterministic pipeline described in Section III-A. The *complete model* adds shelf context to the piecewise-trained candidate CRF.

TABLE I
COMPARISON RESULTS.

Method	Precision	Recall	F1	Exact images
Raw OCR	6.58%	52.39%	11.69%	0/48
OCR + post-processing	82.10%	90.06%	85.90%	9/48
Complete model	94.71%	94.48%	94.59%	21/48

Table I shows that raw OCR has low precision because it returns background text and separates multiline fragments. Post-processing resolves many such cases, while the complete

model further improves F1 and exact-image accuracy through joint correction, grouping, background rejection, and shelf context.

The complete model matches 770 annotations among 813 predictions, giving 43 FP and 45 FN.

C. Ablation Experiments

Four ablations were constructed. Tests are run on Protocol A. *No context* sets the class conflict weight to zero. *Top-1 only* retains the highest-ranked OCR choice at every position. It can still group regions using learned weights. *Full box, no deletion* requires every OCR position in a selected region to be retained. *No join features* removes region count and separator type features while retaining other learned features. Results are shown in Table II.

TABLE II
ABLATION RESULTS. F1 CHANGE IS MEASURED IN PERCENTAGE POINTS RELATIVE TO THE FULL MODEL.

Variant	Precision	Recall	F1	F1 change	Exact images
Complete model	94.71%	94.48%	94.59%	—	21/48
No context	93.87%	93.99%	93.93%	−0.66	19/48
Top-1 only	94.08%	80.00%	86.47%	−8.12	14/48
Full box, no deletion	94.66%	93.50%	94.07%	−0.52	18/48
No join features	94.98%	90.55%	92.71%	−1.88	17/48

Removing context lowers F1 by 0.66 points and exact matches from 21 to 19. Top-1 decoding lowers F1 by 8.12 points, supporting character alternatives. Removing flexible spans lowers it by 0.52 points, consistent with auxiliary text or defects inside a detected region. Without join features, F1 falls by 1.88 points, mainly through lower recall. These ablations support every component.

D. Cross-Library Generalization

Protocol B evaluates transfer ability of the model from the first two libraries to all 60 images and 965 call numbers in the third library. Table III shows that the complete model outperforms deterministic post-processing without using third-library training or validation annotations.

TABLE III
PROTOCOL B CROSS-LIBRARY GENERALIZATION RESULTS.

Method	Precision	Recall	F1	Exact images
Raw OCR	5.06%	45.49%	9.10%	0/60
OCR + post-processing	83.22%	86.84%	84.99%	3/60
Complete model	90.73%	90.26%	90.49%	14/60

The complete model improves F1 over post-processing by 5.50 points and exact images from 3 to 14. This transfer result is below its mixed-library F1, indicating that label and imaging differences across libraries remain challenging.

E. Error Analysis

For 797 of the 815 Protocol A test instances, the exact annotated path was representable under the retained top-10 alternatives for each character, giving an OCR evidence support rate of $797/815 = 97.79\%$. Of the other 18 instances,

13 require a character omitted by the recognizer, three require a character outside the retained alternatives, and two have unusable OCR output or otherwise violate the model assumptions.

Among the 797 supported instances, 782 had a candidate with the annotated string and ordered OCR regions in the generated pool, giving candidate pool recall of $782/797 = 98.12\%$; 15 were lost during candidate searching.

In Protocol A, the complete model matches 770 of 815 annotated call numbers among 813 predictions, leaving 43 false positives and 45 false negatives. In Protocol B, it matches 871 of 965 among 960 predictions, leaving 89 false positives and 94 false negatives.

Previously analyzed examples illustrate the mechanisms. In one success case, the top-1 OCR fragments were 7228 and 2132-2#1, with 2 ranked second at the first position. The model recovered 7228/2132-2#1, whereas post-processing returned 7228/2132-2#1. In a failure case with annotated string J212/4922-1, the initial J was absent from the retained alternatives, so the model could not produce a match.

F. Discussion

The method retains OCR uncertainty and treats character correction, region grouping, and background rejection as one prediction problem. The comparisons across three libraries show gains over fixed post-processing. The ablations support top- k alternatives, flexible spans, learned join features, and the smaller benefit of shelf context.

The method has three main limitations. First, unsupported OCR characters cannot usually be recovered; richer recognizer evidence may help. Second, finite beam search can remove a correct path, so wider adaptive beams may improve recall at added cost. Third, the leading-letter context penalty can disfavor genuine category changes at shelf boundaries. More libraries, label styles, and shelf layouts remain to be evaluated. Per-character coordinates or catalog information could further resolve ambiguous boundaries.

The method has also been integrated into a mobile inventory robot carrying an edge computer and a camera on a vertical rail. Horizontal platform motion and vertical camera motion capture different shelf areas, after which the recognition program produces normalized call numbers for inventory records. This prototype illustrates practical deployment without changing the recognition model described above.

IV. CONCLUSION

This paper presented a context-aware CRF that combines position-level OCR uncertainty, grammar constraints, candidate scoring, shelf context, and global MAP decoding. It achieved 94.59% F1 in Protocol A, compared with 85.90% for deterministic post-processing, and 90.49% on the held-out third library in Protocol B. Future work will study richer OCR evidence, more advanced model factors, and more structured text formats.

REFERENCES

- [1] PaddlePaddle Authors, “PaddleOCR: A practical OCR toolkit,” Software repository, accessed August 12, 2026. [Online]. Available: <https://github.com/PaddlePaddle/PaddleOCR>
- [2] S. S. Tsai, D. M. Chen, H. Chen, C.-H. Hsu, K.-H. Kim, J. P. Singh, and B. Girod, “Combining image and text features: A hybrid approach to mobile book spine recognition,” in *Proceedings of the 19th ACM International Conference on Multimedia*, 2011, pp. 1029–1032.
- [3] X. Yang, D. He, W. Huang, A. G. Ororbia, II, Z. Zhou, D. Kifer, and C. L. Giles, “Smart library: Identifying books on library shelves using supervised deep learning for scene text reading,” in *Proceedings of the 2017 ACM/IEEE Joint Conference on Digital Libraries*, 2017, pp. 1–4.
- [4] A. Llabrés, A. U. Dey, D. Karatzas, and E. Valveny, “Image-text matching for large-scale book collections,” in *Document Analysis Systems*, ser. Lecture Notes in Computer Science, G. Sfikas and G. Retsinas, Eds., vol. 14994. Cham: Springer, 2024, pp. 89–102.
- [5] C. Shi, C. Wang, B. Xiao, Y. Zhang, S. Gao, and Z. Zhang, “Scene text recognition using part-based tree-structured character detection,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2013, pp. 2961–2968.
- [6] S. Tian, Y. Pan, C. Huang, S. Lu, K. Yu, and C. L. Tan, “Text flow: A unified text detection system in natural scene images,” in *Proceedings of the IEEE International Conference on Computer Vision*, 2015, pp. 4651–4659.
- [7] S. Garcia-Bordils, D. Karatzas, and M. Rusiñol, “STEP—towards structured scene-text spotting,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 883–892.
- [8] C. Sutton and A. McCallum, “Piecewise training for undirected models,” in *Proceedings of the Twenty-First Conference on Uncertainty in Artificial Intelligence*. AUAI Press, 2005, pp. 568–575. [Online]. Available: <https://people.cs.umass.edu/mccallum/papers/piecewise-uai05s.pdf>
- [9] D. Koller and N. Friedman, *Probabilistic Graphical Models: Principles and Techniques*. Cambridge, MA: MIT Press, 2009.
- [10] Editorial Committee of the Chinese Library Classification, National Library of China, *Zhongguo Tushuguan Fenleifa (Chinese Library Classification)*, 5th ed. Beijing: National Library of China Publishing House, 2010. [Online]. Available: <https://search.worldcat.org/title/670048842>
- [11] S. Sarawagi and W. W. Cohen, “Semi-Markov conditional random fields for information extraction,” in *Advances in Neural Information Processing Systems*, vol. 17, 2004, pp. 1185–1192.
- [12] X.-D. Zhou, C.-L. Liu, and M. Nakagawa, “Online handwritten Japanese character string recognition using conditional random fields,” in *Proceedings of the 10th International Conference on Document Analysis and Recognition*, 2009, pp. 521–525.
- [13] J. D. Lafferty, A. McCallum, and F. C. N. Pereira, “Conditional random fields: Probabilistic models for segmenting and labeling sequence data,” in *Proceedings of the Eighteenth International Conference on Machine Learning*. Morgan Kaufmann, 2001, pp. 282–289.